TASK 5
ELK SIEM: Custom Correlation Rules and Security Monitoring Dashboard

Project Title: Development and Testing of Custom Correlation Rules in ELK SIEM for Credential Stuffing, DNS Tunneling, and PowerShell Exploitation

Domain: Cybersecurity – Security Operations Center (SOC)

Project Type: Cybersecurity Internship Project
Tools and Technologies: Elasticsearch, Logstash (ELK Stack), Kibana, Winlogbeat, Sysmon, Docker Desktop, Windows 11, Windows Event Logs, PowerShell

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Internship Organization: ELEVANCE SKILLS

Submission Date: OCTOBER

Executive Summary

This project focuses on the development and implementation of a Security Information and Event Management (SIEM) solution using Elasticsearch and Kibana to monitor Windows security events, investigate suspicious activities, and develop custom security detection rules.

The primary objective of the project was to develop and test detection capabilities for multiple failed login attempts, potential DNS tunneling activity, and suspicious PowerShell execution. The project also involved building a centralized security monitoring dashboard to support Security Operations Center (SOC) analysts in identifying and investigating security events.

The SIEM environment was deployed on a Windows 11 system using Docker Desktop to run Elasticsearch and Kibana. Winlogbeat was configured to collect Windows Event Logs and forward them to Elasticsearch for centralized storage and analysis. Microsoft Sysmon was also installed to provide additional endpoint telemetry, including network connection events.

Custom detection rules were configured in Kibana for failed login attempts, DNS tunneling indicators, and PowerShell exploitation indicators. Testing confirmed that the multiple failed login detection rule successfully generated a security alert. The DNS and PowerShell detection rules were configured, although their final detection effectiveness requires further verification.

A custom security monitoring dashboard was developed using Kibana Lens. The dashboard included six visualizations covering failed authentication attempts, targeted usernames, privileged logon events, network connection activity, destination ports, and destination IP addresses.

The project provided practical experience in SIEM deployment, Windows log collection, endpoint monitoring, detection engineering, security event investigation, and dashboard development. It demonstrates how centralized log analysis and custom detection rules can support threat detection and incident investigation in a SOC environment.

1. Introduction

Cybersecurity has become an essential component of modern IT infrastructure due to the increasing number of cyberattacks targeting organizations, networks, and endpoint systems. Security Operations Centers (SOCs) use Security Information and Event Management (SIEM) solutions to collect, monitor, correlate, and analyze security events from multiple sources.

This project focuses on implementing a centralized security monitoring environment using Elasticsearch and Kibana. The system collects Windows security events through Winlogbeat and additional endpoint telemetry through Microsoft Sysmon.

The collected logs are stored in Elasticsearch and analyzed using Kibana to identify suspicious authentication attempts, potential DNS tunneling indicators, and suspicious PowerShell activities.

The project also involves developing custom detection rules and creating an interactive security dashboard to visualize important security events and support SOC investigations.

1.1 Problem Statement

Organizations generate large volumes of security logs every day. Manually reviewing these logs is time-consuming and may result in important security events being overlooked.

Threats such as repeated unauthorized login attempts, suspicious DNS activity, and malicious PowerShell execution can be difficult to identify without centralized monitoring and appropriate detection mechanisms.

Therefore, a centralized SIEM solution is required to collect security events, apply detection rules, and present relevant information through dashboards and alerts.

1.2 Project Objectives

The main objectives of this project are:

1. Deploy Elasticsearch and Kibana using Docker Desktop.
2. Configure Winlogbeat to collect Windows Event Logs.
3. Integrate Microsoft Sysmon for enhanced endpoint monitoring.
4. Develop custom SIEM detection rules for multiple failed login attempts, DNS tunneling indicators, and suspicious PowerShell execution.
5. Generate and analyze security events to evaluate detection functionality.
6. Build a custom Kibana dashboard for visualizing authentication and network activity.
7. Document the implementation, testing procedures, observations, and results.

1.3 Project Scope

The project is implemented in a controlled Windows-based laboratory environment.

The scope includes centralized log collection, endpoint event monitoring, custom detection rule configuration, security alert investigation, and dashboard development.

The project primarily focuses on defensive security monitoring and detection engineering rather than conducting attacks against external systems.

2. Tools and Technologies Used

The ELK SIEM security monitoring project was implemented using open-source security monitoring tools, endpoint logging utilities, and containerization technologies.

2.1 Elasticsearch

Version: 8.13.0

Elasticsearch is a distributed search and analytics engine used to store, index, search, and analyze large volumes of structured and unstructured data.

In this project, Elasticsearch served as the centralized log storage and analysis platform. Security events collected from Windows systems were indexed and made available for investigation and visualization.

2.2 Kibana

Version: 8.13.0

Kibana is a data visualization and analytics platform that integrates with Elasticsearch.

In this project, Kibana was used to:

- Search and investigate Windows security events using Kibana Query Language (KQL).
- Configure custom security detection rules.
- Review security alerts and event information.
- Develop interactive security monitoring dashboards using Kibana Lens.

2.3 Winlogbeat

Version: 8.13.0

Winlogbeat is a lightweight Elastic Beat used to collect Windows Event Logs and forward them to Elasticsearch.

Winlogbeat was configured to collect security-related events from multiple Windows event channels, including:

- Windows Security
- Windows System
- Windows Application
- Windows PowerShell
- Microsoft-Windows-PowerShell/Operational
- Microsoft-Windows-Sysmon/Operational

2.4 Microsoft Sysmon

Version: 15.22

System Monitor (Sysmon) is a Microsoft Sysinternals utility that provides detailed endpoint activity logging.

In this project, Sysmon was installed and configured to collect additional endpoint telemetry, particularly network connection events (Event ID 3).

These events were forwarded through Winlogbeat to Elasticsearch and visualized using Kibana.

2.5 Docker Desktop

Docker Desktop was used to deploy and manage the Elasticsearch and Kibana containers.

Containerization simplified the installation and management of the SIEM environment without requiring a separate dedicated Linux server.

2.6 Windows 11

Windows 11 was used as the host operating system for the

3. System Architecture

3.1 Architecture Overview

The ELK SIEM project follows a centralized security monitoring architecture in which Windows security logs and endpoint activity are collected, forwarded, stored, analyzed, and visualized.

The architecture consists of four primary components:

1. **Windows 11 Endpoint:** Generates Windows Security, System, Application, and PowerShell events. Microsoft Sysmon provides additional endpoint telemetry, including network connection events.
2. **Winlogbeat:** Collects Windows Event Logs and forwards them to Elasticsearch.
3. **Elasticsearch:** Stores and indexes the collected security events for searching, analysis, and detection.
4. **Kibana:** Provides log investigation, custom detection rules, security alerts, and interactive dashboards.

Elasticsearch and Kibana run in Docker containers on the Windows 11 host system.

3.2 Security Log Collection Workflow

The security monitoring workflow operates as follows:

1. Windows 11 generates security and system activity logs.
2. Microsoft Sysmon records additional endpoint activity.
3. Winlogbeat collects events from the configured Windows Event Log channels.
4. Winlogbeat forwards the collected events to Elasticsearch.
5. Elasticsearch indexes the events for centralized analysis.
6. Kibana queries the indexed data to support security investigations, detection rules, and dashboard visualizations.

3.3 Architecture Diagram

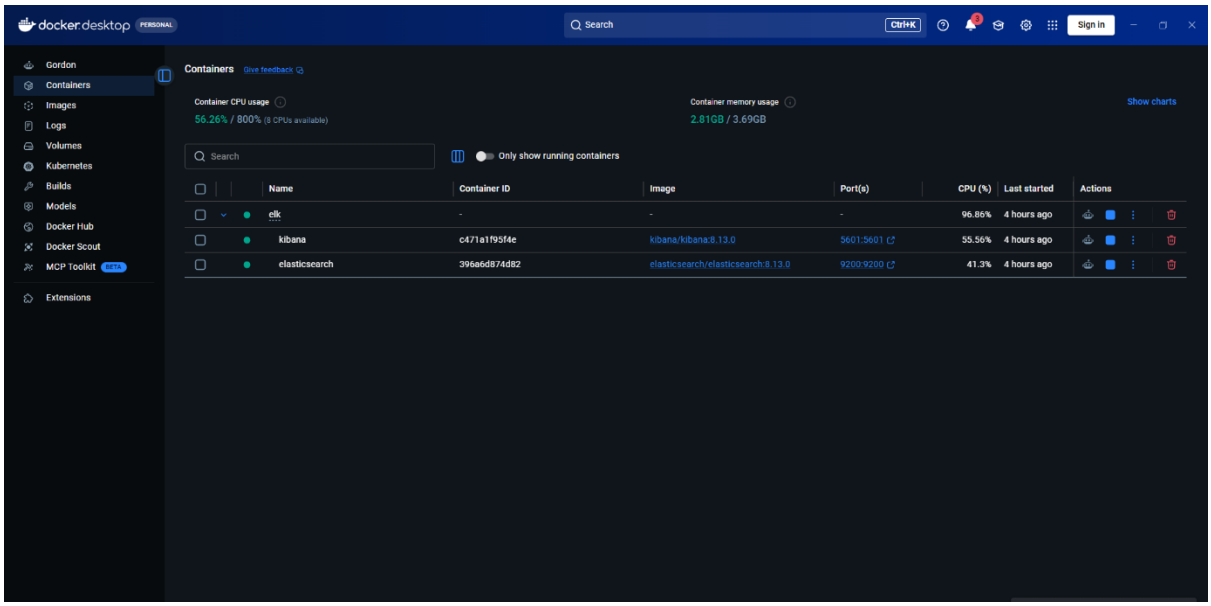


Figure 1 illustrates the logical architecture of the ELK SIEM environment.

4. Installation and Configuration

4.1 Environment Preparation

The SIEM laboratory environment was prepared on a Windows 11 host system. Docker Desktop was used to run Elasticsearch and Kibana as containers.

The following components were configured:

- Host Operating System: Windows 11
- Container Platform: Docker Desktop

- Elasticsearch Version: 8.13.0
- Kibana Version: 8.13.0
- Winlogbeat Version: 8.13.0
- Microsoft Sysmon Version: 15.22

The objective of this setup was to establish centralized security log collection, indexing, investigation, and visualization.

4.2 Elasticsearch Deployment

Elasticsearch was deployed using Docker Desktop and configured to provide centralized storage and indexing for Windows security events.

The Elasticsearch service was exposed through port 9200 on the host system.

The Elasticsearch instance was used to receive logs forwarded by Winlogbeat and make them available for security analysis.

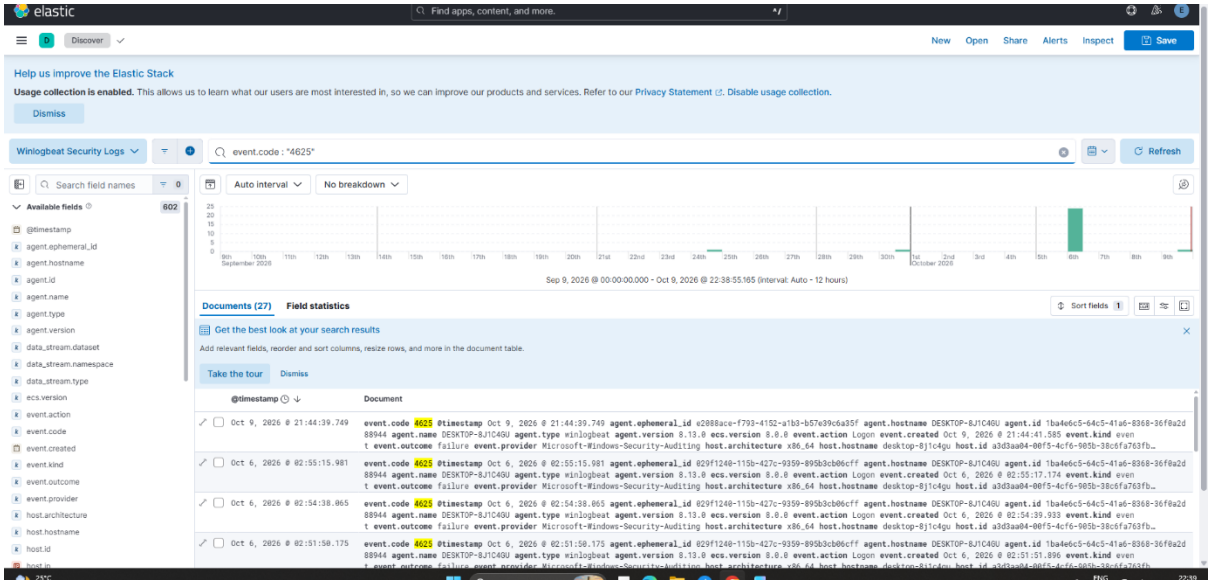


Figure 2: Elasticsearch Container Running in Docker Desktop

4.3 Kibana Deployment

Kibana was deployed as a separate Docker container and connected to Elasticsearch.

The Kibana web interface was accessed through:

http://localhost:5601

Kibana provided an interface for searching collected logs, creating security detection rules, investigating alerts, and developing custom monitoring dashboards.

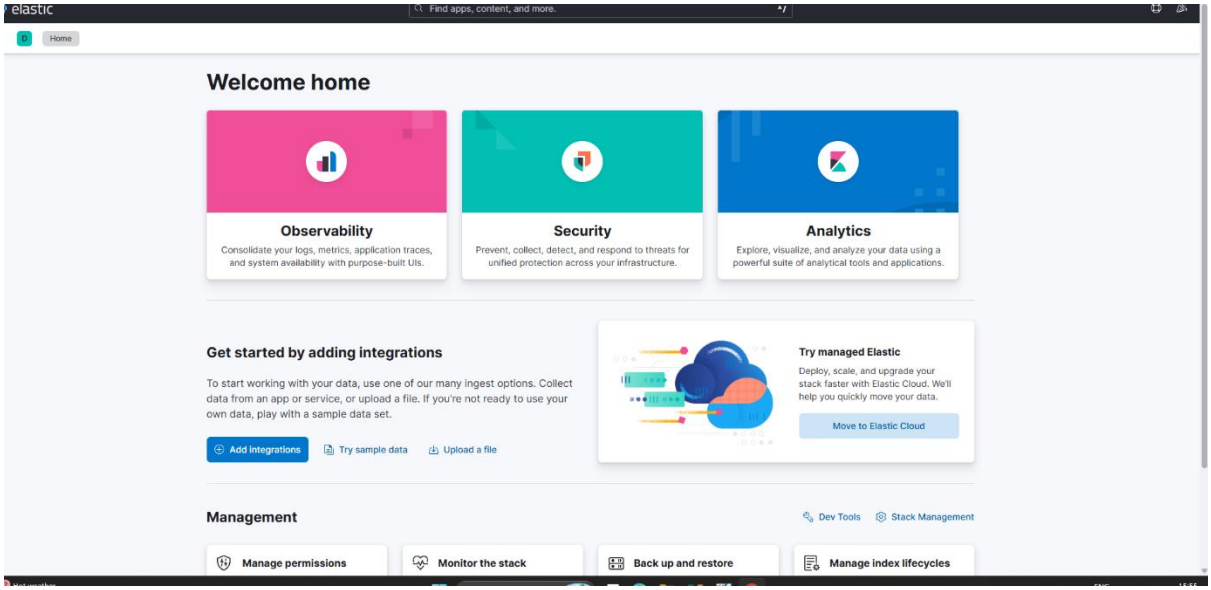


Figure 3: Kibana Web Interface

4.4 Winlogbeat Installation and Configuration

Winlogbeat 8.13.0 was installed and configured to collect Windows Event Logs from the host system.

The Winlogbeat configuration file was located at:

C:\Winlogbeat\winlogbeat-8.13.0-windows-x86_64\winlogbeat.yml

The configuration included Windows Security, System, Application, PowerShell, and Sysmon event channels.

The Winlogbeat service was started to forward collected events to Elasticsearch.

The successful ingestion of Winlogbeat events was verified through Kibana Discover.

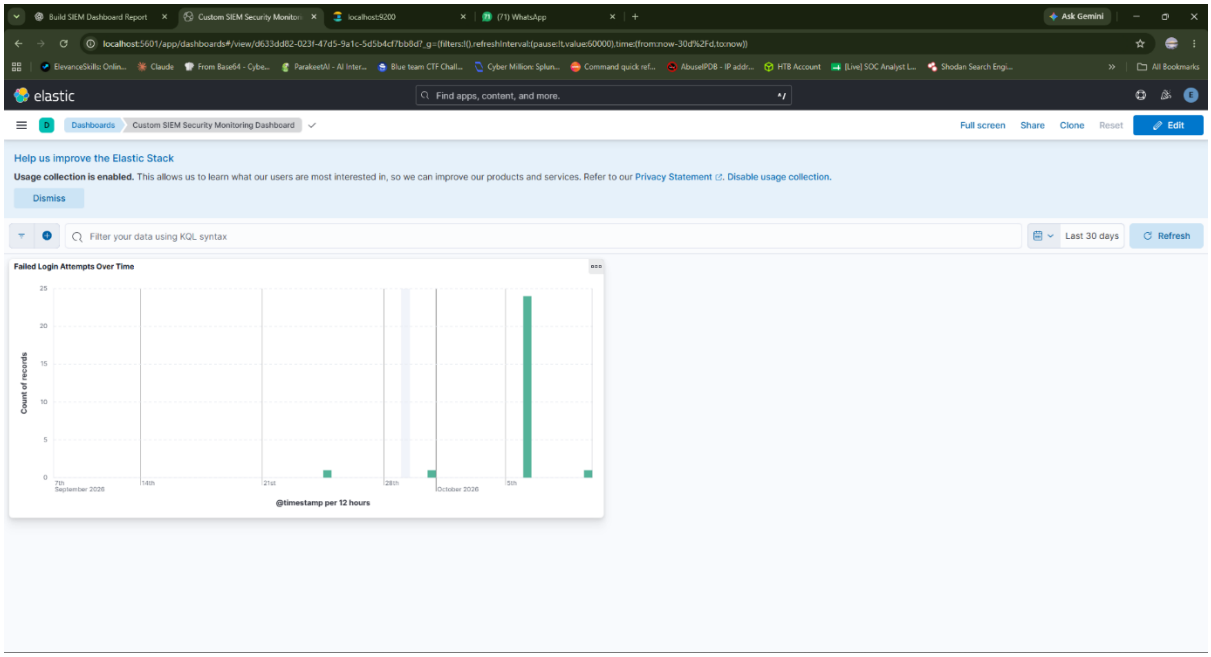


Figure 4: Winlogbeat Configuration and Log Ingestion

4.5 Microsoft Sysmon Installation

Microsoft Sysmon 15.22 was installed to enhance endpoint visibility.

Sysmon was configured to record network connection activity using Event ID 3.

The Sysmon event channel was integrated with Winlogbeat to forward endpoint telemetry to Elasticsearch.

The following Kibana Query Language (KQL) query was used to verify Sysmon network connection events:

winlog.provider_name : "Microsoft-Windows-Sysmon" and event.code : "3"

Matching events were successfully observed in Kibana Discover.

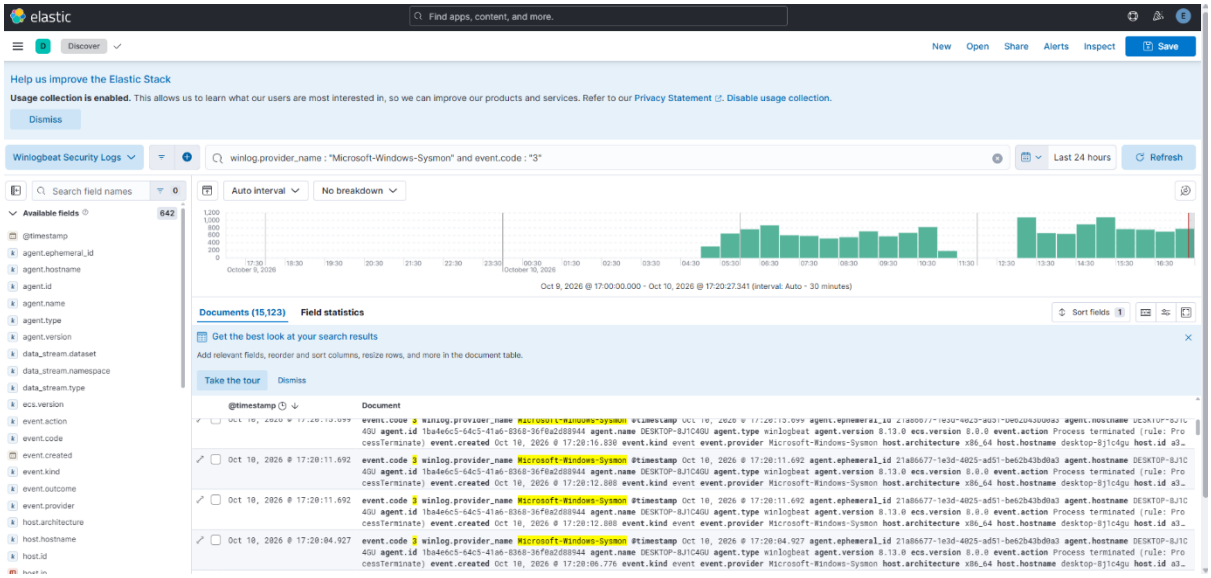


Figure 5: Sysmon Event ID 3 Logs in Kibana Discover

4.6 Verification of Log Collection

After configuring the required components, Kibana Discover was used to verify centralized log collection.

The winlogbeat-* data view was used to investigate indexed Windows events.

The verification confirmed that Windows event logs and Sysmon network connection events were available for searching and visualization.

This established the foundation for developing custom detection rules and security monitoring dashboards.

5. Custom Security Detection Rules

5.1 Overview

Custom security detection rules were configured in Kibana Security to identify potentially suspicious activity within the Windows environment.

The project focused on three detection scenarios:

1. Multiple Failed Login Attempts
2. DNS Tunneling Detection
3. PowerShell Exploitation Detection

These rules were designed to support centralized security monitoring and assist SOC analysts in identifying events that require investigation.

5.2 Rule 1: Multiple Failed Login Attempts

Rule Name: Multiple Failed Login Attempts

Detection Objective: Identify repeated failed Windows authentication attempts that may indicate password guessing, brute-force activity, or credential stuffing.

Relevant Windows Event ID: 4625

Detection Logic: Monitor Windows failed authentication events and identify repeated failures associated with the same user account within a defined time window.

Implementation: A custom detection rule was configured in Kibana Security using collected Windows authentication events.

Testing and Results: Multiple incorrect login attempts were generated in the controlled laboratory environment. The SIEM successfully generated an alert titled "Multiple Failed Login Attempts."

Result: Successful alert generation was confirmed.

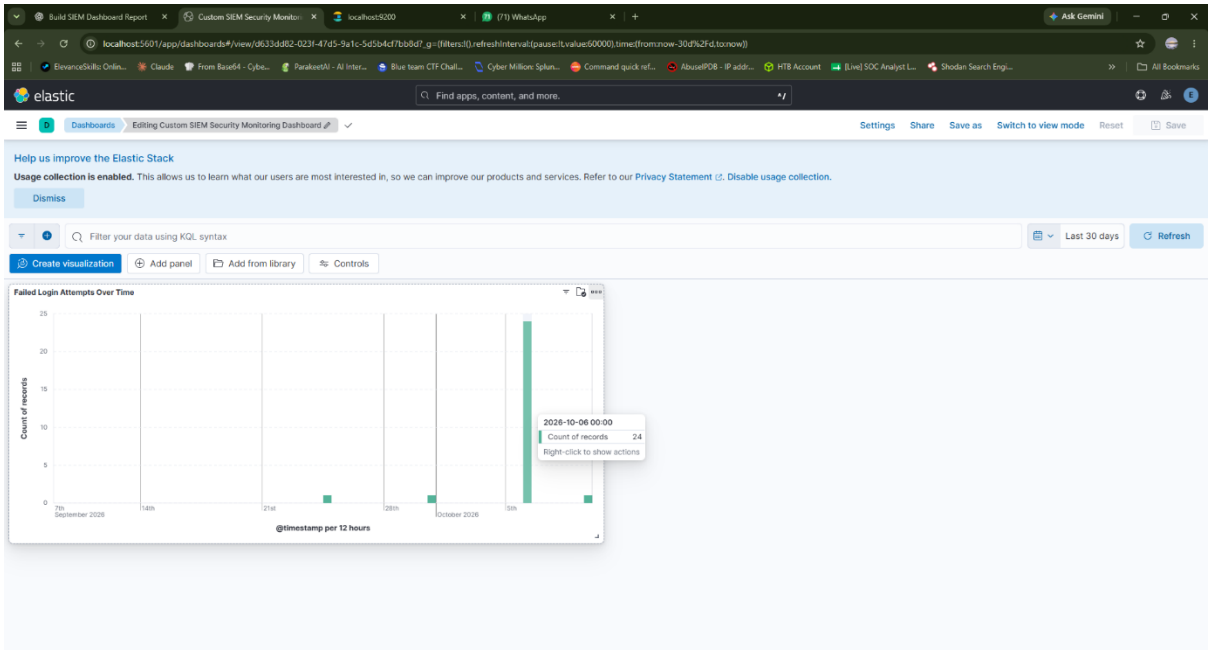


Figure 6: Multiple Failed Login Attempts Detection Rule

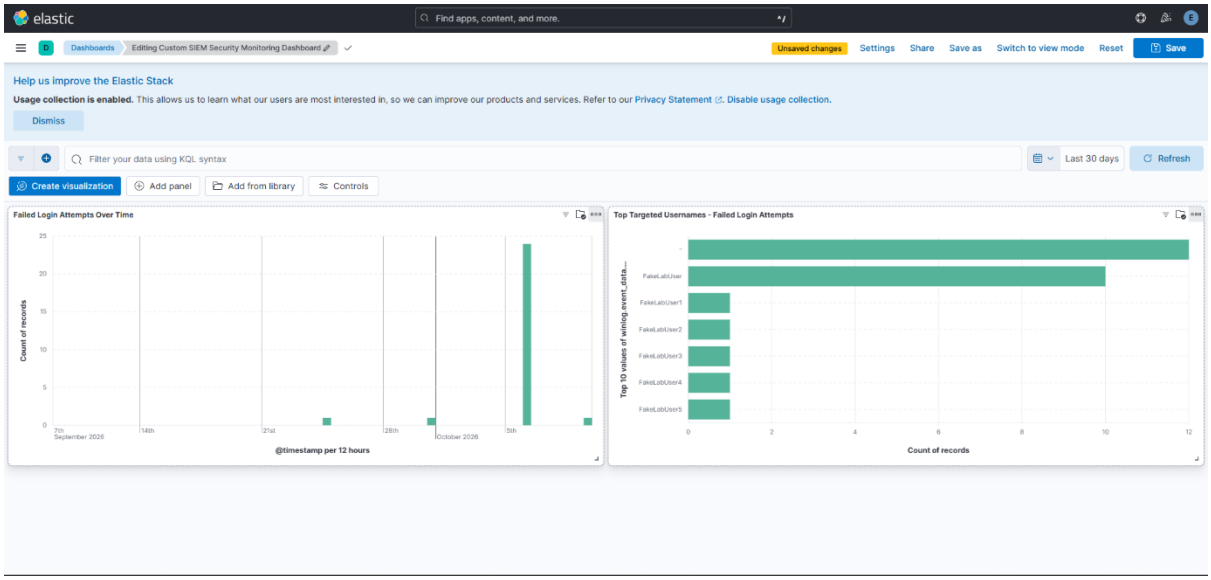


Figure 7: Multiple Failed Login Attempts Security Alert

[Insert screenshot showing the generated alert]

5.3 Rule 2: DNS Tunneling Detection

Rule Name: DNS Tunneling Detection

Detection Objective: Identify potentially suspicious DNS activity that could indicate DNS tunneling or misuse of DNS communication.

Relevant Telemetry: Windows DNS Client Operational logs, including Event IDs 3006 and 3008 observed during laboratory testing.

Detection Approach: Analyze DNS-related events and investigate suspicious query patterns that may require further security analysis.

Implementation: A custom DNS tunneling detection rule was created in Kibana Security. Windows DNS Client Operational logging was enabled to support the collection of DNS-related activity.

Testing and Results: DNS activity was generated using nslookup, and DNS Client Operational events were observed locally. The detection rule was confirmed to exist in Kibana. However, a successful DNS tunneling alert has not yet been independently verified.

Result: Rule configured; alert effectiveness requires further validation.

5.4 Rule 3: PowerShell Exploitation Detection

Rule Name: PowerShell Exploitation Detection

Detection Objective: Identify suspicious PowerShell activity that may indicate unauthorized script execution or potential exploitation.

Relevant Telemetry: Windows PowerShell and Microsoft-Windows-PowerShell/Operational event channels.

Detection Approach: Investigate PowerShell-related events and identify suspicious script execution indicators using collected endpoint telemetry.

Implementation: PowerShell event channels were included in the Winlogbeat configuration, and a custom PowerShell exploitation detection rule was created in Kibana Security.

Testing and Results: PowerShell logging and detection queries were investigated during implementation. The detection rule was confirmed to exist in Kibana. However, a successful alert for this rule has not yet been independently verified.

Result: Rule configured; alert effectiveness requires further validation.

5.5 Detection Rule Summary

Detection Rule	Status	Verified Result
Multiple Failed Login Attempts	Configured and tested	Alert generated
DNS Tunneling Detection	Configured	Alert not yet verified
PowerShell Exploitation Detection	Configured	Alert not yet verified

The results demonstrate successful implementation of custom detection rules and confirmed alert generation for repeated failed authentication attempts. Additional controlled testing is recommended to validate the DNS tunneling and PowerShell detection rules.

6. Attack Simulation, Testing, and Detection Results

6.1 Testing Overview

Controlled security testing was performed to evaluate the log collection and detection capabilities of the ELK SIEM environment.

The testing focused on three security scenarios: repeated failed authentication attempts, DNS activity monitoring, and PowerShell activity investigation.

Additional network connectivity testing was performed to verify Sysmon event collection.

6.2 Test Case 1: Multiple Failed Login Attempts

Objective: Verify whether the SIEM could identify repeated failed Windows authentication attempts.

Procedure:

1. Multiple incorrect login attempts were generated in the controlled Windows environment.
2. Windows Security Event ID 4625 was investigated.
3. The collected authentication events were examined in Kibana.
4. The custom Multiple Failed Login Attempts detection rule was used to identify repeated failures.
5. The Kibana Security alerts interface was reviewed.

Observed Result:

The SIEM successfully generated an alert named "Multiple Failed Login Attempts" with Medium severity.

Outcome: PASS — Successful alert generation was observed.

This test demonstrated the ability to detect repeated authentication failures. It did not independently establish that actual credential stuffing occurred.

6.3 Test Case 2: DNS Activity Monitoring

Objective: Investigate DNS activity and evaluate the available telemetry for potential DNS tunneling detection.

Procedure:

1. Windows DNS Client Operational logging was enabled.
2. DNS lookup activity was generated using nslookup example.com.
3. Windows DNS Client Operational events were investigated.
4. Event IDs 3006 and 3008 were observed locally.
5. The DNS Tunneling Detection rule was configured in Kibana.

Observed Result:

DNS activity was successfully generated and observed in the Windows event logging environment. However, a successful DNS tunneling detection alert was not confirmed.

Outcome: PARTIALLY VERIFIED — Local DNS logging was observed; end-to-end alert detection requires further testing.

6.4 Test Case 3: PowerShell Activity Investigation

Objective: Configure PowerShell event collection and investigate potential suspicious execution indicators.

Procedure:

1. Windows PowerShell logging channels were included in the Winlogbeat configuration.
2. PowerShell-related events were investigated using Kibana queries.
3. A custom PowerShell Exploitation Detection rule was configured.
4. Available logs and detection results were reviewed.

Observed Result:

PowerShell logging channels and a custom detection rule were configured. A successful exploitation-related alert was not independently confirmed.

Outcome: NOT FULLY VERIFIED — Additional test events and alert validation are required.

6.5 Test Case 4: Sysmon Network Connection Monitoring

Objective: Verify that network connection activity could be collected and analyzed through the SIEM.

Procedure:

1. Microsoft Sysmon was installed and configured to collect Event ID 3.
2. Network connectivity was tested using PowerShell.
3. Sysmon network connection events were verified in Windows Event Viewer.
4. Winlogbeat forwarded the Sysmon telemetry to Elasticsearch.
5. Kibana Discover was used to investigate the collected events.

Observed Result:

Sysmon Event ID 3 network connection events were successfully collected and displayed in Kibana.

Outcome: PASS — End-to-end network connection event collection was verified.

6.6 Testing Summary

Test Case	Result
Multiple Failed Login Attempts	PASS
DNS Activity Monitoring	PARTIALLY VERIFIED
PowerShell Activity Investigation	NOT FULLY VERIFIED
Sysmon Network Connection Monitoring	PASS

6.7 Testing Conclusion

The testing demonstrated successful centralized collection of Windows security and Sysmon network connection events, along with confirmed alert generation for repeated failed authentication attempts.

The DNS tunneling and PowerShell detection scenarios require additional controlled testing before their effectiveness can be considered verified.

These findings highlight the importance of validating security detection rules using observable events and generated alerts rather than relying only on successful rule configuration.

7. Custom SIEM Dashboard and Visualization Results

7.1 Dashboard Overview

A custom security monitoring dashboard was developed using Kibana Lens to visualize Windows security events and Sysmon network connection activity.

The dashboard was named **Custom SIEM Security Monitoring Dashboard**.

The objective was to provide SOC analysts with a centralized interface for monitoring authentication failures, privileged logon activity, network connections, and frequently contacted destination ports and IP addresses.

7.2 Dashboard Visualizations

7.2.1 Failed Login Attempts Over Time

This visualization displays the number of failed Windows authentication attempts over time using Windows Security Event ID 4625.

It helps analysts identify periods of increased authentication failures that may require investigation.

7.2.2 Top Targeted Usernames – Failed Login Attempts

This visualization displays user accounts associated with the highest numbers of failed authentication attempts.

It helps analysts identify accounts experiencing repeated unsuccessful login attempts.

7.2.3 Privileged Logon Events Over Time

This visualization monitors Windows Security Event ID 4672, which indicates that special privileges were assigned to a new logon.

It provides visibility into privileged logon activity. These events do not independently prove privilege escalation or malicious behavior.

7.2.4 Network Connections Over Time – Sysmon Event ID 3

This visualization displays network connection activity recorded by Microsoft Sysmon.

It helps analysts identify changes in network connection volume and investigate unusual activity.

7.2.5 Top 10 Destination Ports – Sysmon Event ID 3

This visualization identifies the most frequently contacted destination ports.

It supports network traffic analysis and helps analysts investigate unexpected port usage.

7.2.6 Top 10 Destination IP Addresses – Sysmon Event ID 3

This visualization displays the destination IP addresses most frequently observed in collected Sysmon network connection events.

It helps analysts prioritize network destinations for further investigation.

7.3 Dashboard Results

Six security monitoring visualizations were successfully created and added to the Kibana dashboard.

The dashboard provides a consolidated view of authentication-related activity and endpoint network connections.

These visualizations support security event triage and investigation. Unusual event counts, ports, or IP addresses require additional context before they can be classified as malicious.

7.5 Dashboard Benefits

The custom dashboard provides the following benefits:

- Centralized visibility into Windows security events.
- Easier identification of repeated authentication failures.
- Monitoring of privileged logon events.
- Visibility into endpoint network connection activity.
- Identification of frequently contacted destination ports and IP addresses.
- Support for SOC investigation and event analysis.

8. Challenges and Troubleshooting

8.1 Overview

During the implementation of the ELK SIEM security monitoring environment, several technical challenges were encountered involving Elasticsearch, Kibana, Winlogbeat, Sysmon, and security event collection.

These issues were investigated through service verification, configuration reviews, log analysis, and connectivity testing.

8.2 Elasticsearch Authentication Issue

Problem: Elasticsearch returned an HTTP 401 Unauthorized response during initial connectivity testing.

Cause: The request did not include the required authentication credentials.

Troubleshooting: Elasticsearch authentication requirements and connectivity were investigated.

Outcome: Elasticsearch connectivity was established, allowing further SIEM configuration and log analysis.

8.3 Kibana Server Not Ready

Problem: Kibana initially displayed the message "Kibana server is not ready yet."

Troubleshooting: Docker container status, Elasticsearch connectivity, and Kibana startup behavior were investigated. The Kibana container was restarted during troubleshooting.

Outcome: Kibana became accessible through the web interface.

8.4 Winlogbeat Data Not Appearing in Discover

Problem: Security events were initially not visible in Kibana Discover.

Troubleshooting: Winlogbeat service status, event channel configuration, Elasticsearch indexing, and Kibana data view settings were reviewed.

Outcome: The winlogbeat-* data view became available, and Windows security events were successfully displayed.

8.5 Sysmon Event Collection Issues

Problem: Sysmon network connection events were initially unavailable in Kibana.

Troubleshooting: Sysmon installation, Event ID 3 configuration, Windows Event Viewer, Winlogbeat event channel settings, and service status were investigated.

Outcome: Sysmon Event ID 3 was successfully collected and displayed in Kibana Discover.

8.6 Detection Rule and Alert Verification

Problem: Some detection queries returned no matching results, and the Kibana rule interface initially experienced loading issues.

Troubleshooting: Event fields, KQL filters, rule configuration, and available log data were investigated.

Outcome: Three custom detection rules were created. Successful alert generation was confirmed for repeated failed login attempts. DNS tunneling and PowerShell exploitation alert generation still require further verification.

8.7 Lessons Learned

The troubleshooting process provided practical experience in:

- Diagnosing SIEM connectivity and authentication problems.
- Managing Docker containers and Windows services.
- Configuring Windows event collection.
- Investigating missing or incomplete log data.
- Verifying Sysmon telemetry in Elasticsearch.
- Understanding the difference between configuring a detection rule and validating its effectiveness.

These experiences improved understanding of SIEM deployment, detection engineering, and security monitoring workflows.

9. Conclusion and Future Improvements

9.1 Project Conclusion

The ELK SIEM Custom Correlation Rules and Security Monitoring Dashboard project provided practical experience in implementing a centralized security monitoring environment using Elasticsearch, Kibana, Winlogbeat, and Microsoft Sysmon.

The project successfully established centralized collection and analysis of Windows security events and Sysmon network connection telemetry. Elasticsearch was used for storing and indexing collected events, while Kibana provided security event investigation, detection rule configuration, alert monitoring, and dashboard visualization capabilities.

Three custom detection rules were configured to investigate multiple failed login attempts, potential DNS tunneling activity, and suspicious PowerShell execution.

The Multiple Failed Login Attempts detection rule successfully generated a security alert during controlled testing. The DNS Tunneling Detection and PowerShell Exploitation Detection rules were configured but require additional validation to confirm their effectiveness.

A custom security monitoring dashboard consisting of six visualizations was also developed. These visualizations provided visibility into authentication failures, targeted usernames, privileged logon events, network connections, destination ports, and destination IP addresses.

The project strengthened practical understanding of Security Operations Center (SOC) workflows, log analysis, SIEM deployment, endpoint monitoring, detection engineering, and security event investigation.

Overall, the project demonstrated how centralized log collection, custom detection rules, and security dashboards can support cybersecurity monitoring and incident investigation.

9.2 Future Improvements

The following enhancements could improve the SIEM environment:

1. **Advanced Detection Rules:** Develop and validate additional correlation rules for suspicious authentication activity, malware indicators, and lateral movement.
2. **DNS Tunneling Validation:** Generate controlled DNS tunneling-like test events and evaluate the detection rule using measurable indicators.

3. **PowerShell Detection Validation:** Enable appropriate PowerShell logging and perform controlled tests to verify detection of suspicious script execution.
4. **Threat Intelligence Integration:** Enrich security events with trusted threat intelligence to help analysts investigate suspicious IP addresses and domains.
5. **MITRE ATT&CK Mapping:** Map validated detection rules to relevant MITRE ATT&CK techniques to improve detection coverage analysis.
6. **Automated Alert Notifications:** Configure notification connectors to support timely investigation of security alerts.
7. **Additional Endpoint Monitoring:** Integrate more Windows or Linux endpoints to expand monitoring coverage.
8. **Detection Performance Evaluation:** Measure detection accuracy, false positives, and alert response times using repeatable test scenarios.

9.3 Final Learning Outcomes

Through this project, practical knowledge was gained in:

- Deploying Elasticsearch and Kibana using Docker Desktop.
- Configuring Winlogbeat for Windows Event Log collection.
- Integrating Microsoft Sysmon with the SIEM environment.
- Investigating Windows security events using Kibana Query Language.
- Creating and testing custom security detection rules.
- Analyzing security alerts and endpoint network connections.
- Developing security monitoring dashboards using Kibana Lens.
- Troubleshooting SIEM connectivity and log ingestion problems.
- Documenting cybersecurity implementation and testing results.

The knowledge and experience gained from this project provide a foundation for further learning in SOC operations, threat detection, incident response, and security engineering.

10. References

The following official documentation and technical resources are relevant to the implementation of this project.

1. **Elastic — Elasticsearch Reference Documentation**
   https://www.elastic.co/guide/en/elasticsearch/reference/8.13/index.html
2. **Elastic — Kibana Documentation**
   https://www.elastic.co/guide/en/kibana/8.13/index.html
3. **Elastic — Winlogbeat Documentation**
   https://www.elastic.co/guide/en/beats/winlogbeat/8.13/index.html
4. **Microsoft — Sysmon Documentation**
   https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
5. **Microsoft — Windows Security Auditing Event 4625**
   https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/event-4625
6. **Docker — Docker Desktop Documentation**
   https://docs.docker.com/desktop/
7. **MITRE ATT&CK Framework**
   https://attack.mitre.org/

11. Appendix

Appendix A: Project Screenshots

The following screenshots document the implementation, configuration, and testing of the ELK SIEM security monitoring environment.

Figure	Description
Figure 1	ELK SIEM System Architecture
Figure 2	Elasticsearch Container in Docker Desktop
Figure 3	Kibana Web Interface
Figure 4	Winlogbeat Configuration and Log Ingestion
Figure 5	Sysmon Event ID 3 Logs in Kibana
Figure 6	Multiple Failed Login Attempts Detection Rule
Figure 7	Failed Login Security Alert
Figure 8	DNS Tunneling Detection Rule

Figure 9	PowerShell Exploitation Detection Rule
Figure 10	Custom ELK SIEM Security Monitoring Dashboard

Appendix B: Security Events Investigated

Event ID	Description
4625	Failed Windows Logon
4672	Special Privileges Assigned to New Logon
3 (Sysmon)	Network Connection
3006 / 3008	DNS Client Operational Events Observed During Testing

Appendix C: Project Deliverables

1. ELK SIEM implementation and configuration.
2. Winlogbeat and Sysmon log collection.
3. Three custom security detection rules.
4. Verified failed login detection alert.
5. Custom security monitoring dashboard with six visualizations.
6. Testing and troubleshooting documentation.
7. Final cybersecurity internship project report.